

PRINCIPLES OF PHARMACOLOGY

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Terms used to describe pharmacology

Pharmacokinetics – how substances are distributed through the body

Pharmacodynamics – interactions of the substance with a receptor

WHAT IS PHARMACOLOGY?



the scientific study of the actions of **drugs** and their effects on a living organism

WHAT IS A DRUG?

Illicit substances
(drugs of abuse)



Medicines
• plant-based
• synthetic

OTC substances
• Aspirin
• Omega-3 oils

THERE ARE SPECIALIZED FIELDS OF PHARMACOLOGY

Neuro-pharmacology

- drug-induced changes in nervous system function

Psycho-pharmacology

- drug-induced changes in mood, thinking, and behavior

Neuropsychopharmacology

- drug-induced changes in the nervous system that influence behavior

WHAT'S THE DIFFERENCE BETWEEN

Drug action

is a specific molecular change

drug binds to its target (receptor)

Drug effect

is a more widespread / systemic change

e.g., in physiology or behavior

Drug actions can be very distant from the ultimate effect

WHAT'S THE DIFFERENCE BETWEEN

Therapeutic effects

Drug effect

Desirable

Side effects

Drug effect

Undesirable
Annoying, distressing, dangerous

Drug actions occur at many different target sites

THE PHARMACOLOGY OF PROZAC

a commonly used SSRI antidepressant medication.

- Binds to serotonin transporter
- Blocks the reuptake of serotonin

Drug action

Drug effects

- Alters the physiology of serotonin system
- Therapeutic effect – relief from depression
- Side effects – headache, insomnia, sexual dysfunction

WHAT ELSE CAN DRUGS DO?

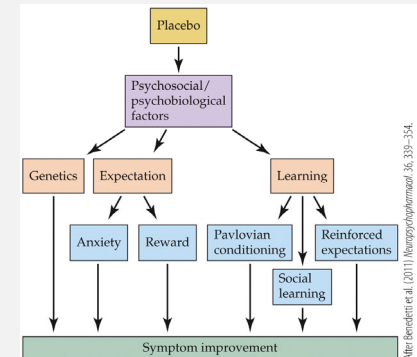
Specific drug effects

- result from interactions between a drug and its target (pharmacodynamics)

Nonspecific drug effects

- may also affect outcome of drug use
- include mood, expectations, perceptions, attitudes

2 EXAMPLES OF NONSPECIFIC DRUG EFFECTS



PHARMACO-KINETICS (PK)

- Mechanisms involved in delivering a drug to its target, where it can have a pharmacological effect.
- Administration
- Absorption
- Distribution
- Elimination
- Metabolism & Excretion

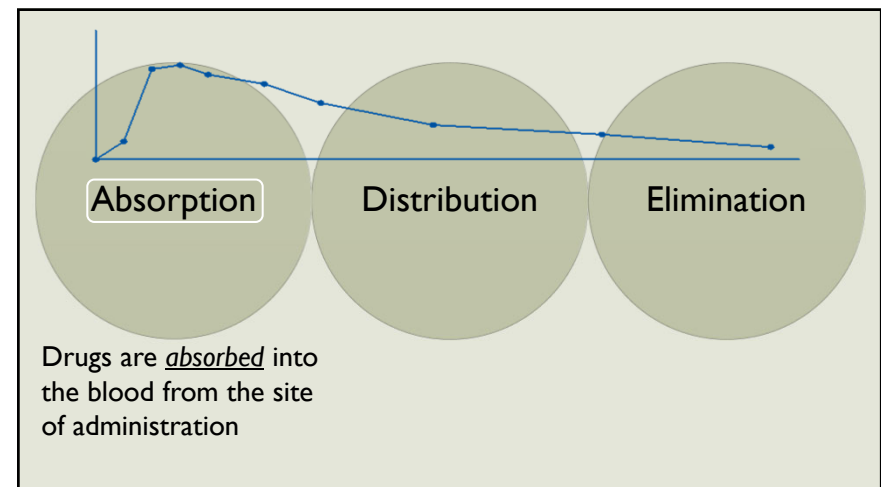
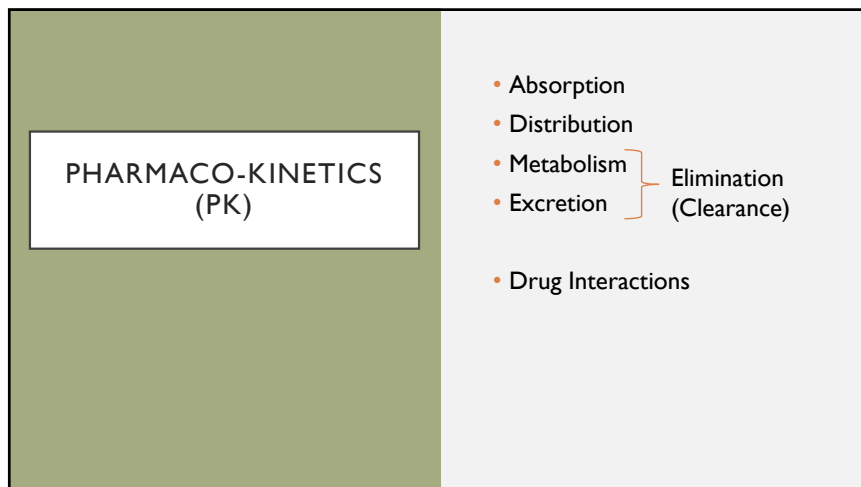
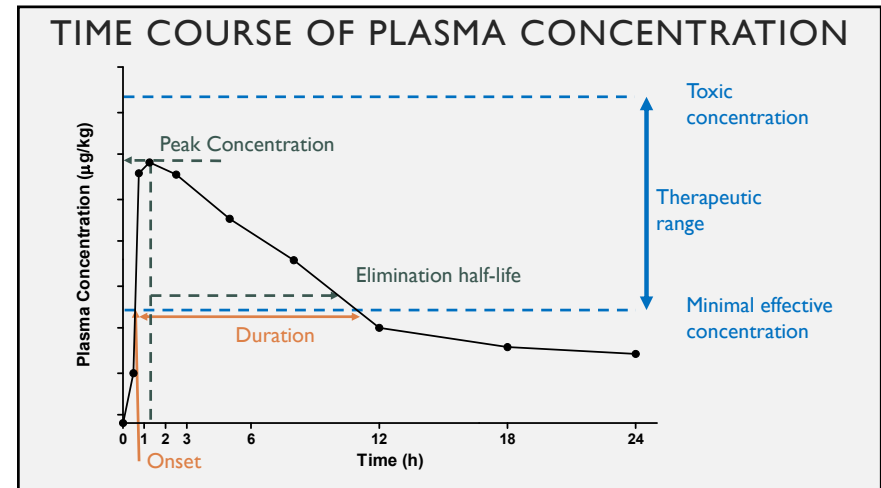
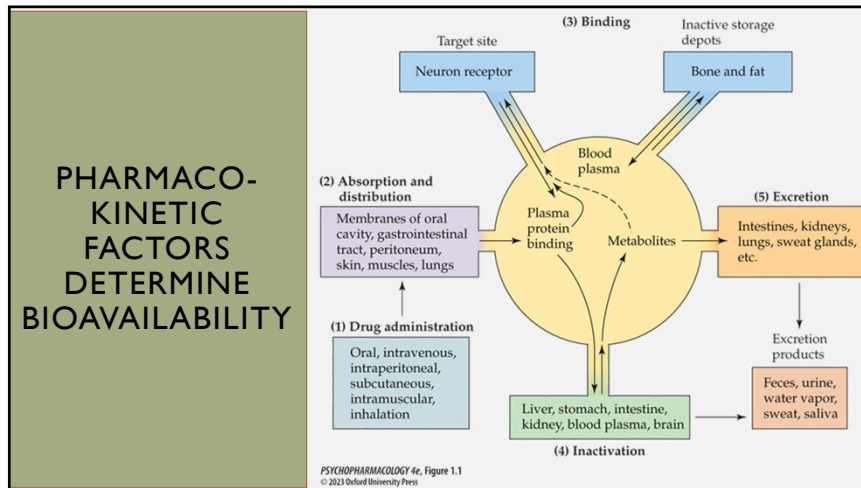
BIOAVAILABILITY

is the amount of drug in the blood that is free to act on a specific target

depends on several factors, including

- absorption into the blood by route of administration
 - affected by lipid solubility, ionization, selective barriers
- drug distribution to target sites
 - plasma proteins, depot binding
- elimination / clearance mechanisms
 - For a drug to be bioactive it must be unchanged (not metabolized) or active metabolite

Ch 01 – Pharmacokinetics & Pharmacodynamics



Ch 01 – Pharmacokinetics & Pharmacodynamics



PHARMACOKINETICS

- Absorption is the movement of drug from its site of administration into the blood.
- Oral delivery requires absorption through the GI tract.
- Drugs must survive this “first-pass” metabolism
- Once in the blood, a drug can reach its active site or bind to inactive sites in blood (e.g., albumin) muscle or fat.

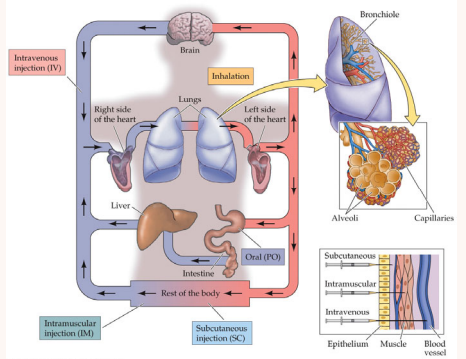
4 COMMON ROUTES OF ADMINISTRATION

Oral ingestion	• tablet, capsule, liquid
Injection	• intramuscular, intravenous
Inhalation	• smoking, inhalers, nebulizers
Through mucous membranes	• sublingual, intranasal, rectal suppository, transdermal

ROUTE OF ADMINISTRATION

affects both

- bioavailability
- drug half-life



PSYCHOPHARMACOLOGY 4e, Figure 1.2
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ABSORPTION IS AFFECTED BY MULTIPLE FACTORS

- Lipid solubility
- Ionization
- Diffusion across membranes

ABSORPTION IS AFFECTED BY MULTIPLE FACTORS

Lipid solubility

- Cell membranes are comprised of phospholipid bilayers
- Substances must be lipid-soluble to pass through the membranes by passive diffusion
- Movement is down a concentration gradient

Ionization

Diffusion across membranes

MEMBRANE LIPID BILAYER

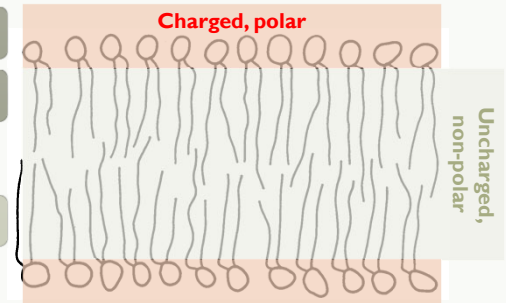
Head group

Phosphate group

- carry ionic charge
- polar, hydrophilic

2 fatty acid tails

- do not carry charge
- non-polar, hydrophobic



ABSORPTION IS AFFECTED BY MULTIPLE FACTORS

Lipid solubility

Ionization

- Most psychoactive drugs are weak acids or weak bases that are ionized in water
- Ionization depends on pH of solution
- Blood, CSF pH $\approx$ 7.4; stomach pH $\approx$ 1 - 3
- Example: aspirin

Diffusion across membranes

Aspirin is a weak acid

- pKa $\sim$ 3.5

1- Mostly non-ionized in stomach

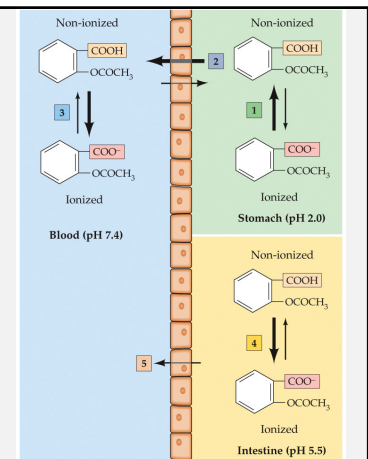
- 2- Passes through the stomach walls into the blood

3- Mostly ionized in the blood

- trapped

4- Mostly ionized in the intestines

- 5- Absorption is slowed



PSYCHOPHARMACOLOGY 4e, Figure 1.6
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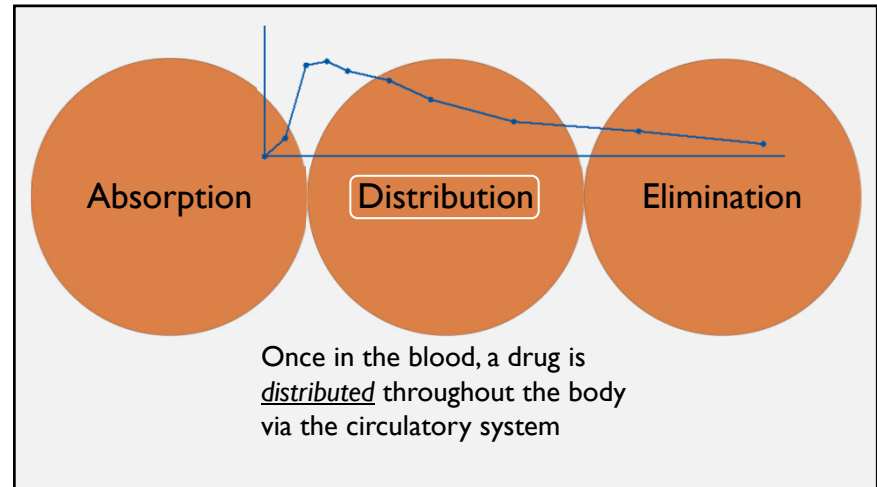
ABSORPTION IS AFFECTED BY MULTIPLE FACTORS

Lipid solubility

Ionization

Diffusion across membranes

- Blood-brain barrier
- Placental barrier



THE AMOUNT OF DRUG AT A TARGET SITE IS A FRACTION OF THE TOTAL DOSE ADMINISTERED.

Drug depots (or depot binding)

- plasma proteins, especially albumin
- muscle & fat
- non-selective (low affinity) binding
- drug displacement (e.g., phenytoin displaced by aspirin)

THE AMOUNT OF DRUG AT A TARGET SITE IS A FRACTION OF THE TOTAL DOSE ADMINISTERED.

Drug depots (or depot binding)

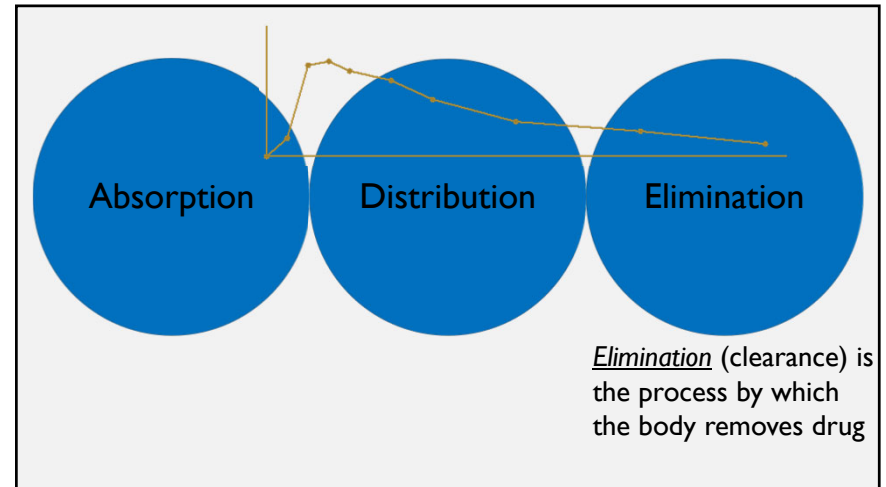
- Affect drug bioavailability & drug effects, including side-effects
 - delays onset
 - lowers intensity
 - prolongs duration

Ch 01 – Pharmacokinetics & Pharmacodynamics

Illustrate (draw / sketch) the factors that affect

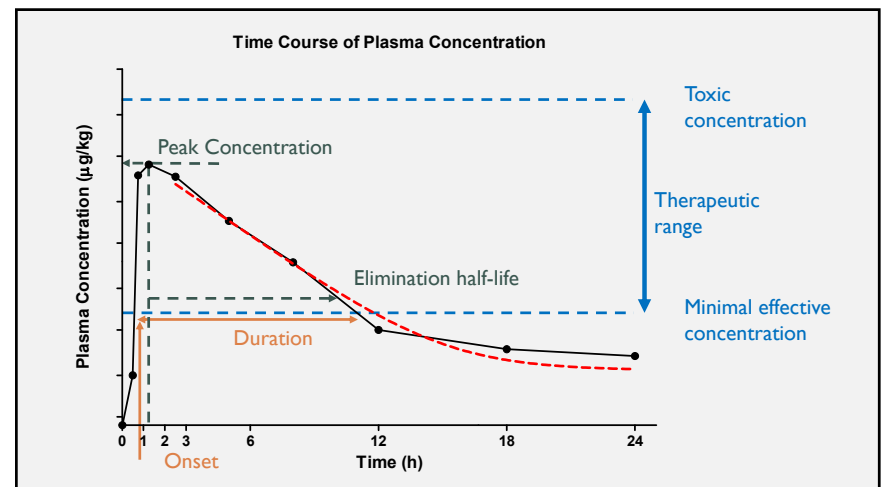
- absorption and
 - distribution of a substance,
- comparing 2 routes of administration: one enteral, and one parenteral.

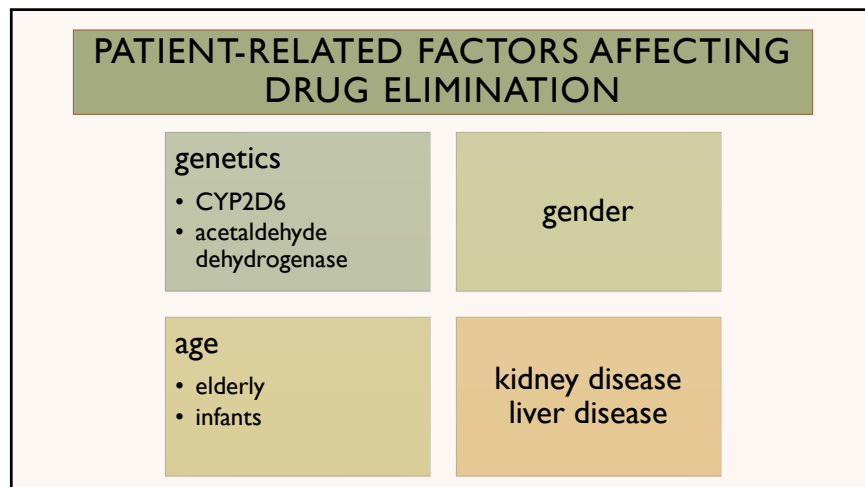
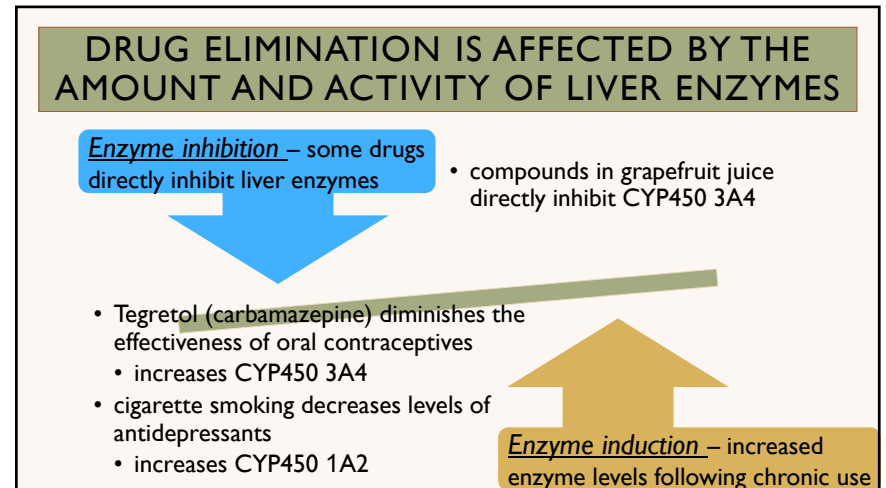
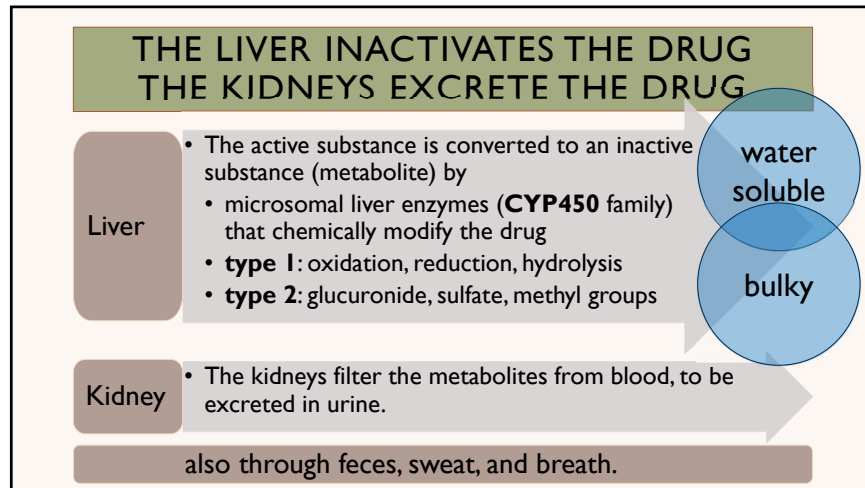
Relate any changes to absorption and distribution to drug effects such as intensity and/or duration of drug effects.



PSYCHOACTIVE DRUGS ARE COMMONLY METABOLIZED BY THE LIVER AND EXCRETED BY THE KIDNEY

- Elimination is usually exponential, follows 1st-order kinetics
- rates depend on drug concentration in the blood,
- half-life ($t_{1/2}$) – amount of time to clear 50% of drug
- and on the blood flow into the metabolizing organ
- clearance rate is an important factor when considering dosing schedule.





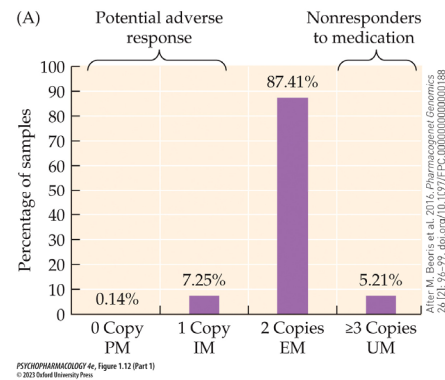
Compare and contrast Type 1 and Type 2 metabolism

Make 2 columns.

- Column 1 – how are they similar,
- Column 2 – how are they different?

Ch 01 – Pharmacokinetics & Pharmacodynamics

FOUR
GENETIC
POPULATIONS
BASED ON
THE NUMBER
OF
FUNCTIONAL
CYP2D6
GENES



PHARMACO-
KINETIC
FACTORS
DETERMINE
BIOAVAILABILITY

